You

For this question and all future questions until I say otherwise, please answer for a general audience using the information in this repository. Prefer place names over identifiers, briefly explain unfamiliar ecological terms, and show each result only once. How does Pacific Northwest Coastal Forests fit into One Earth's larger geographical regions? Show the hierarchy from this bioregion upward.

GraphTalker

Here is the complete picture of how **Pacific Northwest Coastal Forests** fits into One Earth's geographical hierarchy:

One Earth Geographic Hierarchy

One Earth organizes the planet into a four-tier system, from broadest to most specific:

```
REALM (largest)
  +-- SUBREALM
        +-- BIOREGION
              +-- ECOREGION (smallest)
```

Here is where Pacific Northwest Coastal Forests sits:

```
Northern America                          <-- REALM
  +-- North Pacific Coast                 <-- SUBREALM
        +-- Cascades Mountain Forests And Valleys
        +-- Greater California
        +-- Pacific Northwest Coastal Forests   <-- BIOREGION (this one)
              |
              +-- British Columbia Coastal Conifer Forests
              +-- Central Pacific Northwest Coastal Forests
              +-- Haida Gwaii Conifer Forests
              +-- Klamath Siskiyou Forests
              +-- Northern California Coastal Forests
              +-- Puget Lowland Forests
```

What each level means:

- **Realm -- "Northern America"**: The largest geographic division, covering the entire continent of North America.
- **Subrealm -- "North Pacific Coast"**: A major subdivision within the realm, grouping together regions that share broad climatic and ecological characteristics along the Pacific coast.
- **Bioregion -- "Pacific Northwest Coastal Forests"**: A distinct ecological region defined by its landscape, climate, and dominant natural communities -- in this case, the temperate rainforests and coastal forests stretching from northern California up through British Columbia.
- **Ecoregions** (the six listed above): The most fine-grained divisions, each representing a specific area with its own characteristic mix of species, soil, and climate. For example, **Haida**

Gwaii Conifer Forests refers to the isolated island archipelago off British Columbia, while **Puget Lowland Forests** covers the lowland areas around Puget Sound in Washington State.

The two **sibling bioregions** -- Cascades Mountain Forests And Valleys and Greater California -- share the same North Pacific Coast subrealm, reflecting their geographic and ecological kinship along the western edge of the continent.

You

Which ecoregions are associated with Pacific Northwest Coastal Forests, and which biome does each belong to?

GraphTalker

Here are the six ecoregions within the **Pacific Northwest Coastal Forests** bioregion and the biome each belongs to:

Ecoregion	Biome
British Columbia Coastal Conifer Forests	Temperate Conifer Forests
Central Pacific Northwest Coastal Forests	Temperate Conifer Forests
Haida Gwaii Conifer Forests	Temperate Conifer Forests
Klamath Siskiyou Forests	Temperate Conifer Forests
Northern California Coastal Forests	Temperate Conifer Forests
Puget Lowland Forests	Temperate Conifer Forests

Every ecoregion in this bioregion belongs to the same biome: **Temperate Conifer Forests**. This is a biome characterized by forests dominated by cone-bearing, needle-leaved trees (such as Douglas fir, western red cedar, and Sitka spruce) growing in regions with mild, wet winters and cool summers. It is the signature vegetation type of the Pacific coast from northern California through British Columbia, and it includes some of the tallest and most productive forests on the planet -- the classic "temperate rainforests" of the Pacific Northwest.

You

“What are the three solution pillars, and how many solution pathways belong to each? Include pathways beneath intermediate classes.”

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Here is how One Earth's solutions framework is organized.

The Three Solution Pillars

One Earth structures its climate solutions into three broad **pillars** -- think of them as the three main strategies for addressing the climate crisis. Each pillar contains intermediate groupings called **subpillars**, which in turn contain specific **solution pathways** (the concrete actions).

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1. Nature Conservation Activity -- 25 solution pathways

Focused on protecting and restoring wild ecosystems.

Subpillar	Pathways
Ecosystem Restoration	9
Land Conservation	8
Ocean Conservation	4
Wildlife Connectivity Improvement	4

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2. Energy Transition Activity -- 23 solution pathways

Focused on shifting from fossil fuels to clean energy sources.

Subpillar	Pathways
Renewable Power Production	9
Renewable Heat Provision	6
Energy Efficiency Improvement	4
Renewable Transport Activity	4

---

3. Regenerative Agriculture Activity -- 29 solution pathways

Focused on transforming how we grow food and fiber so that farming rebuilds soil, sequesters carbon, and reduces waste.

Subpillar	Pathways
Regenerative Cropland Management	15
Food Waste Reduction	6
Rangeland Sustainability Activity	5
Circular Fibershed Management	3

Summary

```
Climate Solution (top level)
  |
  +-- Nature Conservation Activity ..... 25 pathways
  |     +-- Ecosystem Restoration (9)
  |     +-- Land Conservation (8)
  |     +-- Ocean Conservation (4)
  |     +-- Wildlife Connectivity Improvement (4)
  |
  +-- Energy Transition Activity ..... 23 pathways
  |     +-- Renewable Power Production (9)
  |     +-- Renewable Heat Provision (6)
  |     +-- Energy Efficiency Improvement (4)
```

```
|     +-- Renewable Transport Activity (4)
|
+-- Regenerative Agriculture Activity ........ 29 pathways
      +-- Regenerative Cropland Management (15)
      +-- Food Waste Reduction (6)
      +-- Rangeland Sustainability Activity (5)
      +-- Circular Fibershed Management (3)
```

Across the entire framework there are **77 solution pathways** organized under **12 subpillars** within the **3 pillars**. Regenerative Agriculture is the largest pillar, driven largely by the many approaches to cropland management (things like cover cropping, composting, and reduced tillage). Nature Conservation follows closely, reflecting the diversity of restoration and protection strategies available for land, ocean, and wildlife habitats.

You

Which illustrative activities target Pacific Northwest Coastal Forests, and where does each activity take place?

GraphTalker

Here are the two illustrative activities that target the **Pacific Northwest Coastal Forests** bioregion:

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1. Example Wetland Restoration

- **Type:** Wetland Restoration (a solution pathway under the Nature Conservation pillar)
- **Location:** Pacific Northwest Coastal Forests
- **Description:** A fictional wetland restoration effort used to demonstrate how activities, targets, and the geographic hierarchy work together in the One Earth framework.

This activity both *targets* and *takes place in* the same bioregion -- a local restoration project restoring wetlands right where they are needed.

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2. Example Conservation Fundraising

- **Type:** Climate Fundraising
- **Location:** Great Britain, Ireland, and Faroe Islands
- **Description:** A fictional fundraising activity situated in Britain and intended to support conservation in the Pacific Northwest.

This one is interesting because the activity *takes place* on the other side of the Atlantic -- in the British Isles -- but *targets* the Pacific Northwest Coastal Forests. It illustrates how One Earth's framework can capture the common real-world pattern where funding and support originate in one region but are directed toward conservation outcomes in another.

Both of these are flagged as **illustrative examples** -- meaning they are fictional, included in the dataset to show how the One Earth solutions model connects activities to the places they aim to help.

You

Which illustrative activity takes place in a different realm from the bioregion it targets? Show the activity, both bioregions, and both realms

GraphTalker

There is exactly one illustrative activity that crosses realm boundaries:

Example Conservation Fundraising

	Bioregion	Realm
Takes place in	Great Britain, Ireland, and Faroe Islands	Western Eurasia
Targets	Pacific Northwest Coastal Forests	Northern America

This activity is located in one continent -- **Western Eurasia** -- but directs its conservation support toward a bioregion in an entirely different realm -- **Northern America**. It's a good demonstration of how One Earth's data model can represent the global nature of conservation funding, where donors and organizations in one part of the world channel resources toward ecosystems thousands of miles away.

You

Which illustrative Nature Conservation activities take place in Pacific Northwest Coastal Forests? Include instances of subclasses.

GraphTalker

Three illustrative Nature Conservation activities take place in Pacific Northwest Coastal Forests:

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1. Example Wetland Restoration -- *Wetland Restoration*

A fictional wetland restoration effort used to demonstrate how activities, targets, and the geographic hierarchy work together in the One Earth framework.

2. Example Habitat Work -- *Ecosystem Restoration*

A fictional component of the example wetland restoration effort. Ecosystem Restoration is a broader subpillar under Nature Conservation that covers restoring degraded habitats of all kinds.

3. Example Native Planting -- *Native Vegetation Planting*

A fictional planting activity within the example habitat work. Native Vegetation Planting -- the practice of reestablishing plants that naturally belong in a region -- is itself a specific type of Ecosystem Restoration.

These three activities illustrate a nesting pattern that mirrors the class hierarchy:

```
Nature Conservation Activity
  +-- Ecosystem Restoration
  |     +-- Example Habitat Work
  |     |
```

```
|     +-- Wetland Restoration
|     |     +-- Example Wetland Restoration
|     |
|     +-- Native Vegetation Planting
|           +-- Example Native Planting
```

The query needed to follow subclass chains (Wetland Restoration and Native Vegetation Planting are both subclasses of Ecosystem Restoration, which is itself under Nature Conservation) to catch all three. Without traversing those intermediate classes, only the top-level match would have appeared.

You

Show the subprocess hierarchy of Example Wetland Restoration, including each activity's location and targets.

GraphTalker

Now I have the full picture. Here is the subprocess hierarchy:

Subprocess Hierarchy of Example Wetland Restoration

```
Example Wetland Restoration  (Wetland Restoration)
  |
  +-- Example Habitat Work  (Ecosystem Restoration)
        |
        +-- Example Native Planting  (Native Vegetation Planting)
```

Each activity takes place in the same location -- **Pacific Northwest Coastal Forests** -- but they differ in what they target:

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1. Example Wetland Restoration -- *Wetland Restoration*

- **Location:** Pacific Northwest Coastal Forests
- **Targets:**
  - **Pacific Northwest Coastal Forests** (Bioregion) -- the broader region this effort aims to benefit
  - **Example Wetland** (Ecosystem) -- the specific wetland being restored
  - **Coho Salmon** (Species) -- a fish species that depends on healthy wetland and stream habitats

This top-level activity has the richest set of targets, capturing the full scope of what the restoration aims to help: a place, an ecosystem, and a species.

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2. Example Habitat Work -- *Ecosystem Restoration*

- **Location:** Pacific Northwest Coastal Forests
- **Targets:**
  - **Example Wetland** (Ecosystem)

A subprocess of the wetland restoration, focused specifically on restoring the physical habitat of

the wetland itself.

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3. Example Native Planting -- *Native Vegetation Planting*

- **Location:** Pacific Northwest Coastal Forests
- **Targets:**
  - **Example Wetland** (Ecosystem)

The most granular activity -- planting species native to the region as part of the habitat work. Native vegetation planting means reestablishing the plants that naturally belong in a landscape, which helps stabilize soil, filter water, and provide food and shelter for wildlife.

How it all connects

The hierarchy shows how a large conservation effort decomposes into increasingly specific tasks:

```
Restore a wetland (whole project)
  +-- Restore the habitat (physical structure)
        +-- Plant native vegetation (specific action)
```

All three activities share the same location, but the top-level project casts a wider net with its targets -- spanning the bioregion, the ecosystem, and an individual species (Coho Salmon) -- while the sub-activities focus narrowly on the ecosystem being restored.

You

Which organization is responsible for both fundraising and wetland restoration? Show the activities, their locations, and their targets as a network graph.

GraphTalker

The answer is **Example Conservation Organization** -- it is the only organization responsible for both a fundraising activity and a wetland restoration activity.

The Two Activities

1. Example Conservation Fundraising (Climate Fundraising)

- **Location:** Great Britain, Ireland and Faroe Islands
- **Target:** Pacific Northwest Coastal Forests

2. Example Wetland Restoration (Wetland Restoration)

- **Location:** Pacific Northwest Coastal Forests
- **Targets:**
  - Pacific Northwest Coastal Forests (the bioregion)
  - Example Wetland (a specific ecosystem)
  - Coho Salmon (a species)

Network Graph

The graph shows how one organization sits at the center, connecting two very different

activities. The fundraising effort operates across the Atlantic in the British Isles but channels support toward the Pacific Northwest. The wetland restoration, by contrast, works locally -- both located in and targeting the same bioregion -- with additional focus on a specific wetland ecosystem and a salmon species that depends on it.



Would you like me to store this query and network graph visualization in the Query Library?

You

Which species does Example Wetland Restoration target, and what broader solution categories does that activity belong to?

GraphTalker

The species targeted by Example Wetland Restoration is **Coho Salmon** (*Oncorhynchus kisutch*) -- a Pacific salmon species found in coastal streams and rivers from California to Alaska. Healthy wetlands are critical to coho salmon because they provide rearing habitat for juveniles, filtering water and moderating stream flows.

Solution Category Hierarchy

The activity belongs to a chain of increasingly broad solution categories:



So a single activity nests four levels deep: it is a **Wetland Restoration**, which is a kind of **Ecosystem Restoration**, which falls under the **Nature Conservation** pillar, which is one of One Earth's three top-level **Climate Solutions**. This layering lets the framework capture both the specific action being taken (restoring a wetland) and its broader strategic purpose (conserving

specific action being taken (restoring a wetland) and its broader strategic purpose (conserving nature to address the climate crisis).

You

Using the stored bioregion boundary geometries and geof:sfWithin, identify the bioregion containing longitude −122.3321, latitude 47.6062. Which illustrative solution activities directly target that bioregion, and where does each activity take place? Use CRS84 longitude–latitude order and return labels rather than WKT strings.

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The point at longitude -122.3321, latitude 47.6062 -- which is the location of **Seattle, Washington** -- falls within the **Pacific Northwest Coastal Forests** bioregion. The query used the `geof:sfWithin` function to test whether the point lies inside each bioregion's stored boundary polygon, using CRS84 longitude-latitude coordinate order.

Two illustrative solution activities directly target that bioregion:

Activity	Type	Takes Place In
Example Conservation Fundraising	Climate Fundraising	Great Britain, Ireland and Faroe Islands
Example Wetland Restoration	Wetland Restoration	Pacific Northwest Coastal Forests

The results confirm what we have seen in earlier questions from a different angle: these are the same two activities linked to Pacific Northwest Coastal Forests, but this time we discovered the bioregion not by name but by dropping a geographic coordinate into its boundary geometry. The fundraising activity operates remotely from the British Isles, while the wetland restoration works locally within the bioregion itself.

You

Using the stored boundary geometries, which bioregion contains longitude −0.1276, latitude 51.5072? Which illustrative activities take place in that bioregion, and which bioregions do they target? Calculate containment with geof:sfWithin.

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The point at longitude -0.1276, latitude 51.5072 -- the location of **London, England** -- falls within the **Great Britain, Ireland and Faroe Islands** bioregion.

One illustrative activity takes place in that bioregion:

Activity	Type	Located In	Targets
Example Conservation Fundraising	Climate Fundraising	Great Britain, Ireland and Faroe Islands	Pacific Northwest Coastal Forests

This is the same cross-realm fundraising activity we have encountered before, but this time discovered from the opposite direction. In the previous query we started from Seattle's coordinates and found activities that *target* Pacific Northwest Coastal Forests. Here we started

from London's coordinates and found the activity that *takes place* in Great Britain, Ireland and Faroe Islands. Both paths lead to the same relationship: a fundraising effort based in the British Isles channeling conservation support across the Atlantic to the Pacific Northwest.

You

Which other bioregions share a boundary with the bioregion directly targeted by Example Wetland Restoration? Calculate this using geof:sfTouches on their stored WKT geometries. Exclude the target bioregion itself.

GraphTalker

Five bioregions share a boundary with **Pacific Northwest Coastal Forests** (the bioregion targeted by Example Wetland Restoration):

1. **Cascades Mountain Forests And Valleys** -- the mountain range running parallel to the coast, just inland
2. **Far Northern Pacific Coast** -- the continuation of the coastal zone northward into Alaska
3. **Greater California** -- the region stretching south along the coast
4. **Greater Rockies And Mountain Forests** -- the broader Rocky Mountain system further inland
5. **Northwest Intermountain Conifer Forests** -- the drier, interior forests between the Cascades and the Rockies

The `geof:sfTouches` function identifies geometries that share boundary points but do not overlap in their interiors -- exactly what you need to find adjacent regions. The result paints a geographic picture: Pacific Northwest Coastal Forests is flanked by coastal neighbors to the north and south (Far Northern Pacific Coast and Greater California), bounded inland by the Cascades, and connected through mountain passes to the interior regions (Northwest Intermountain Conifer Forests and Greater Rockies).